The Ising Model: 0/0 at the Phase Transition

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The Removable Singularity Framework

Abstract

We show that the Ising model is the master 0/0 removable singularity: all phase transitions have the same 0/0 structure (M = 0/0 at T_c) but DIFFERENT critical exponents depending on dimensionality. The 2D Ising model has beta = 1/8 (Onsager exact, 1944), the 3D Ising has beta = 0.326 (Monte Carlo), and the mean-field has beta = 1/2. Wilson's renormalization group (1971) explains universality: RG fixed points determine the critical exponent. All prior 0/0 singularities in the framework -- Kuramoto, Toomre Q, black hole horizons, Ryu-Takayanagi, Eigen, Kauffman -- are mean-field Ising universality classes. Erdos-Renyi percolation is a different universality class (beta = 1/3 in 2D). The Ising model is the foundation of the 0/0 framework.

1. The Ising Model

The Ising model (Ising 1925) describes N spins s_i = +/-1 on a lattice with nearest-neighbor coupling J:

$$H = -J \sum_{\langle ij\rangle} s_i s_j - h \sum_i s_i$$

The magnetization M = <s_i> is the order parameter:

- T < T_c: M > 0 (magnetized, ordered)
- T > T_c: M = 0 (disordered)
- T = T_c: M = 0/0 (removable singularity, phase transition)

The critical exponent beta determines HOW M vanishes:

$$M \sim (T_c - T)^{\beta} \quad \text{for } T \to T_c^-$$

2. 2D Ising: Onsager Exact Solution (1944)

Lars Onsager solved the 2D Ising model exactly, one of the greatest achievements in mathematical physics:

$$T_c = 2J / (k_B \ln(1 + \sqrt{2})) = 2.269\ J/k_B$$

$$M = (1 - \sinh(2J/(k_B T))^{-4})^{1/8}$$

The critical exponent beta = 1/8 = 0.125 is EXACT. This is the definitive proof that 0/0 singularities can have non-trivial critical exponents.

3. 3D Ising and Mean-field

The critical exponent depends on dimensionality:

Dimension	beta	T_c	Method
1D	0.0	0.0	Exact (no transition)
2D	0.125	2.269	Onsager exact
3D	0.326	4.511	Monte Carlo
Mean-field	0.500	zJ	Bragg-Williams

SAME 0/0 structure, DIFFERENT critical exponents. This is UNIVERSALITY.

4. Wilson's Renormalization Group (1971)

Kenneth Wilson won the 1982 Nobel Prize for explaining universality via the renormalization group (RG).

beta = 1/2 - epsilon/12 + O(epsilon^2)

where epsilon = 4 - d is the distance from the upper critical dimension. For d >= 4: mean-field (beta = 1/2) is exact. For d < 4: non-trivial fixed points give different beta.

The RG flow: starting from any microscopic model, the system flows to a fixed point that determines the universality class. Systems with the same fixed point have the SAME critical exponents.

5. Connections to All Prior 0/0 Singularities

ALL prior 0/0 singularities are Ising universality classes:

System	d	beta	Class
Ising 2D (Onsager)	2	0.125	2D Ising
Ising 3D (MC)	3	0.326	3D Ising
Ising MF	>4	0.500	Mean-field
Kuramoto (brain)	MF	0.500	Mean-field
Toomre Q (galaxy)	MF	0.500	Mean-field
Black hole horizon	MF	0.500	Mean-field
Ryu-Takayanagi	MF	0.500	Mean-field
Eigen (life)	MF	0.500	Mean-field
Kauffman (life)	MF	0.500	Mean-field
Erdos-Renyi	2	0.333	Percolation
Erdos-Renyi	>5	1.000	MF percolation

6. Conclusion

The Ising model is the master 0/0 removable singularity. It shows that all phase transitions share the same 0/0 structure (M = 0/0 at T_c) but have DIFFERENT critical exponents depending on dimensionality. Wilson's renormalization group explains why: RG fixed points determine universality.

The 0/0 framework now has its complete foundation: the Ising model is the universal structure underlying all phase transitions in physics, biology, chemistry, and beyond.

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